

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-  
a07c4284c19178092.jsonl`  
- SHA-256: `1e0b2a6064ac95ba23d132364d589eee771d473ab3f91bf14638f0bdd912587c`  
- Source modified: `2026-06-06T17:30:52+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:30:14.121Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** <https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/>

****Title:**** Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System (PMC, 2025)

****Found via:**** broad/primary search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:17.131Z)

I'll fetch the source page and extract the relevant claims.

3. user (2026-06-06T17:30:32.796Z)

Acoustic Detection of Tennis Ball Hits Using Machine Learning

(1) Method - Acoustic Signal Detection

The study uses audio signals to detect ball impacts and timing:

> "classifies the sound of the ball's impact on the racket and the ground to measure the time between them"

The system performs peak detection on acoustic waveforms, distinguishing between ball impact sounds and ground rebound sounds from background noise.

(2) Machine Learning Models and Accuracy Metrics

****For impact detection:**** XGBoost classifier achieved high performance

> "accuracy across each fold ranged between 0.96 and 0.99, with a median value of 0.98"

****For rebound detection:**** Ensemble approach combining XGBoost, SVM, and MLP

> "accuracy for each fold ranged from 0.95 to 0.98, resulting in a median value of 0.97"

****Overall system accuracy on-court:****

> "the ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%"

For groundstrokes specifically: detected impacts in 96.25% and 81.25% of cases; rebounds in 86.25% and 70.83% of cases across two groups.

(3) Hardware and Sensors

Commercial action camera microphone:

> "stereo audio recording from the built-in microphone set at 48 kHz"

Specifically, a GoPro 11 camera was used. Processing occurred on a notebook computer.

(4) Publication Date

> "Received 2024 Nov 13; Revised 2025 Jan 21; Accepted 2025 Jan 23"

Published January 27, 2025.

(5) Temporal Precision Details

Timing measurement accuracy compared to video analysis:

> "no statistically significant differences in the group medians" ($G1 ? = 9 \text{ ms}$, $p = 0.415$; $G2 ? = 9 \text{ ms}$, $p = 0.943$)

Video analysis temporal resolution: "time absolute error of the camera can be assumed as equal to a frame time ($?t = 4 \text{ ms}$)"

(6) Dataset Information

The training dataset contained:

> "1400 audio samples previously acquired" with "509 sounds for the class noise, 442 for rebound, and 448 for the impact"

****Public availability:**** The dataset is not public. Authors state:

> "The raw data supporting the conclusions of this article will be made available by the corresponding author upon request"

4. assistant (2026-06-06T17:30:35.878Z)

I have all the information needed. Let me assess and structure the findings.

This is a peer-reviewed primary research article from PMC (PubMed Central), published 2025. It directly bears on the audio-based hit detection family in the research question.

5. user (2026-06-06T17:30:43.657Z)

Structured output provided successfully

6. assistant (2026-06-06T17:30:52.186Z)

I've extracted and structured the key claims from this PMC source. Summary of my assessment:

****Source quality: primary**** ? This is a peer-reviewed primary research article hosted on PubMed Central (NIH), with full received/revised/accepted dates, an experimental methodology, and reported metrics.

****Relevance to the research question:**** Directly relevant to the audio-based hit detection family. It provides quantitative evidence that acoustic ball-impact detection is viable on commodity hardware (a single camera microphone + laptop) with near-frame timing precision (~9 ms), which is exactly the kind of fully-local, low-compute signal the badminton pipeline could exploit from its existing audio track.

****Key takeaways for the pipeline:****

- Audio hit detection in isolation is very accurate (median 0.98 per-fold; >95% overall), but end-to-end on-court accuracy drops to 85% ? a realistic caution that real-world noise (crowd, multiple courts) degrades performance.
- It runs locally on a notebook from a 48 kHz mic ? no cloud, no GPU bottleneck, fitting the local-first/permissive constraint.
- Caveats: it is tennis (not badminton), trained on only ~1400 samples, and the dataset is not public (on-request only), so the labels/weights aren't directly reusable commercially.

The 5 falsifiable claims with supporting quotes have been returned via the StructuredOutput tool.